

ENGR 1451/2451: Exploratory Data Science

Simple Linear Regression I

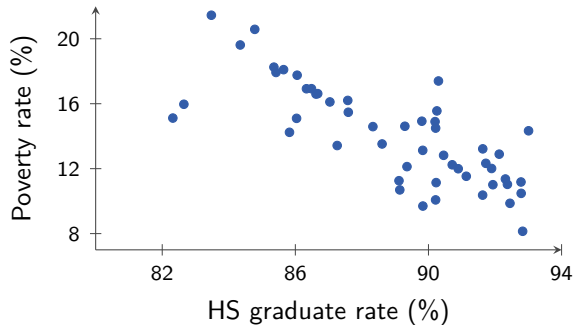
Prof. Paul W. Leu
University of Pittsburgh

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Today: Fit, Interpret, and Check the Limits

- Use **one predictor** to fit a straight-line average trend.
- Let **Python obtain the slope and intercept**.
- Interpret coefficients and predictions in the variables' units.
- Check the prediction direction, data range, and meaning of the observations.

Example: Poverty and HS Graduates



What is one observation?

One of the **50 US states or DC**.

Variables

x : HS graduate rate (%).

y : residents living in poverty (%).

These are **region-level percentages**, not individual people.

2017 state/DC averages computed from county-level data in `county_complete.csv`.

Simple Linear Regression: Our Scope

One predictor, one quantitative response

Use a straight line to estimate how the **average response** varies with x .

- **Simple:** one predictor.
- **Here:** a straight line in x .
- **Fitting method:** ordinary least squares, with an intercept.

Fitted line

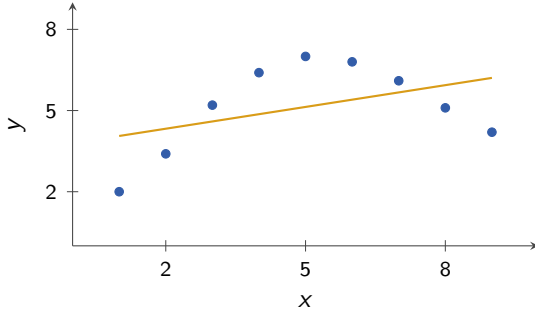
$$\hat{y} = b_0 + b_1x$$

- b_0 : intercept.
- b_1 : slope.
- $\hat{y}$: fitted or predicted response; it need not equal observed y .

Regression is broader than this lesson

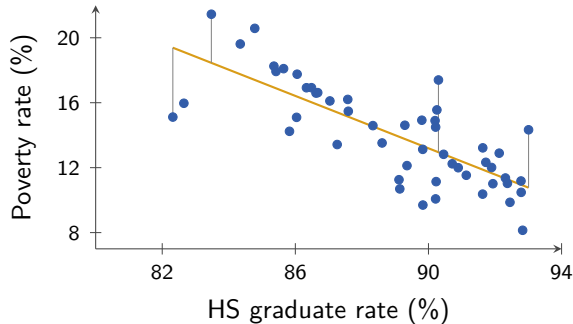
Other regression models can use more predictors or describe curved relationships.

A Line Can Be Fitted Even When It Is a Poor Summary



- This average pattern is curved.
- Python will still return a slope and intercept.

How Does Python Choose the Line?



Ordinary least squares

Choose the slope and intercept that make the **sum of squared vertical residuals** as small as possible.

$$\text{SSE} = \sum_i (y_i - \hat{y}_i)^2$$

- Vertical differences: we are predicting y from x .
- Squaring makes large misses count more.

Python: Obtain the Slope and Intercept

Python

```
slope, intercept = np.polyfit(x, y, deg=1)  
  
print(f"Slope: {slope:.6f}")  
print(f"Intercept: {intercept:.6f}")
```

Output

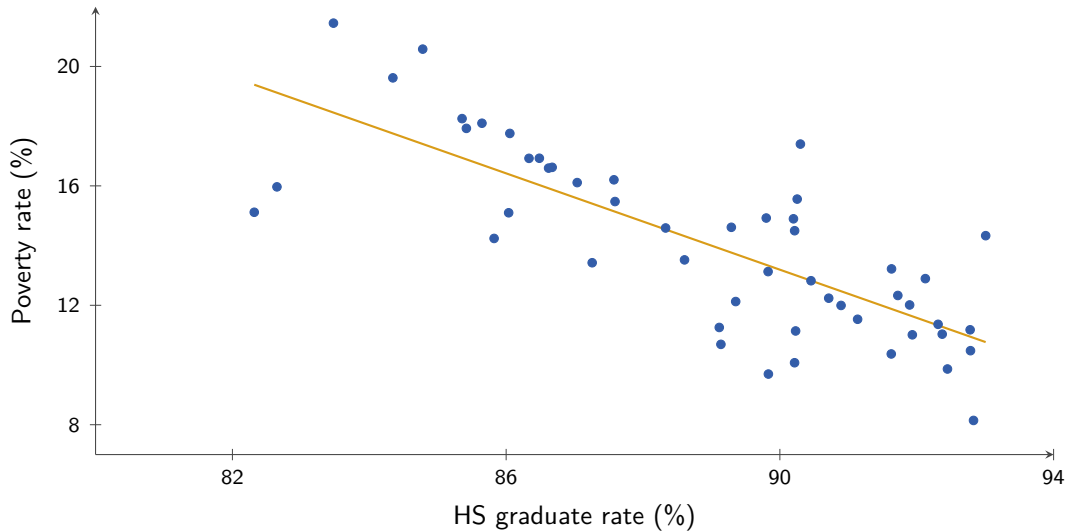
```
Slope: -0.806625  
Intercept: 85.789160
```

Read the fitted equation

$$\widehat{\text{poverty}} \approx 85.789 - 0.806625 \text{ hs_grad}$$

- deg=1: fit a straight line.
- Return order: **slope, then intercept**.
- Python computes the coefficients from the paired data.

Draw the Line Over the Observations



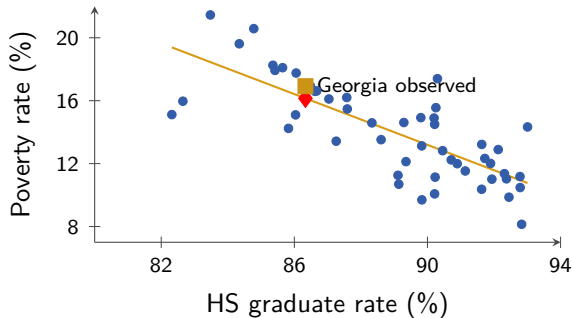
Quick Check: Which Slope Interpretation Works?

The fitted slope is approximately -0.807 .

Choose the defensible statement

1. Raising a region's HS graduate rate by one point will cause poverty to fall by 0.807 points.
2. A one-point higher HS graduate rate is associated with a 0.807-point lower predicted poverty rate.
3. Each additional HS graduate reduces the poverty rate by 0.807%.

A Prediction Is Not Every Observed Outcome



Georgia

HS graduate rate: **86.3%**

Predicted poverty: **16.15%**

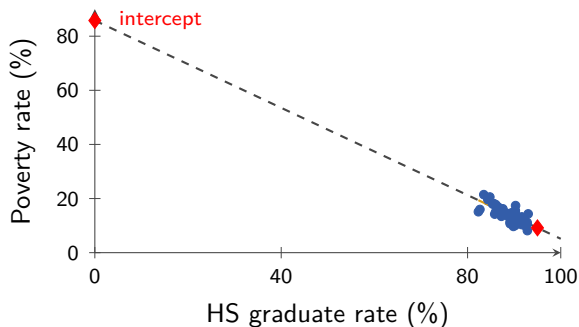
Observed poverty: **16.9%**

The line estimates an average trend. It does not reproduce each state's outcome.

Residual

Observed minus predicted: about **+0.77** percentage points.

Check Whether a Prediction Is Outside the Data



Notebook prediction table

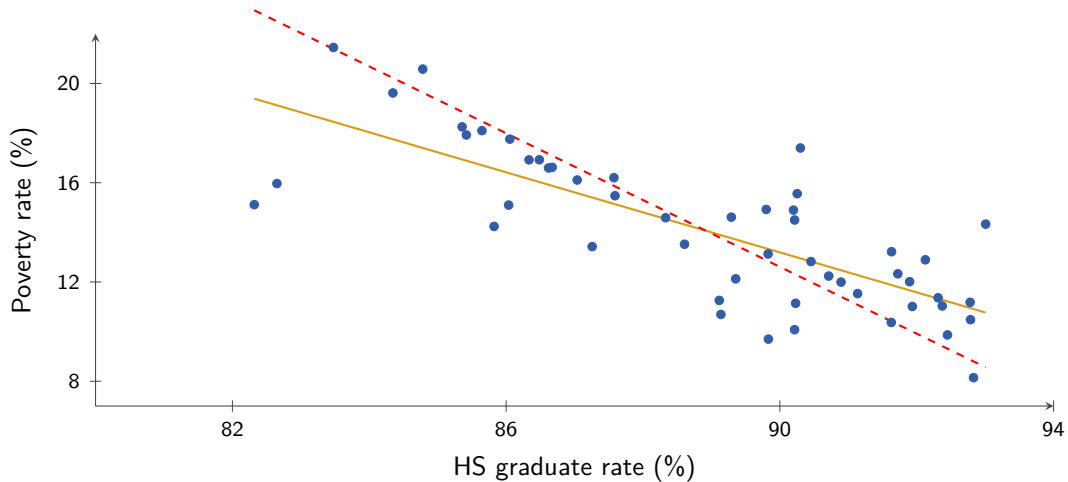
hs_grad	predicted_poverty	inside
0.00	85.79	False
86.33	16.15	True
95.00	9.16	False

Observed HS-rate range

82.32% to 93.01%

Python will compute a value outside the observed range; that does not validate the extrapolation.

Reverse Prediction Requires a New Fit



Gold: poverty-on-HS fit. Dashed red: reverse fit, shown on the same axes after solving for poverty.

Association Is Not Intervention

Observational study

The slope describes average differences among groups already in the data. It does **not** automatically predict what would happen if we intervened to change x .

Controlled experiment

If x is assigned by the investigator and the mechanism is understood, the slope can be interpreted as an average causal response, within the experimental range.

An Association Is Not an Intervention

What the model describes

Among these regions, higher HS graduate rates tend to accompany lower poverty rates.

What it does not establish

The amount poverty would change if an intervention increased a region's HS graduate rate.

Keep the unit of observation

These are **regional aggregates**.
The line does not tell us the poverty risk of an individual graduate.

Other characteristics of the regions may contribute to the observed relationship.

Regression to the Mean

A related phenomenon

With imperfect correlation between comparable repeated measurements, groups selected for extreme first results can be less extreme on the second measurement.

This is a statement about group averages, not every individual.

Example from the earlier deck

For comparable standardized tests with $r = 0.50$, a first score 2 SDs above average has a fitted prediction only 1 SD above average on the second test.

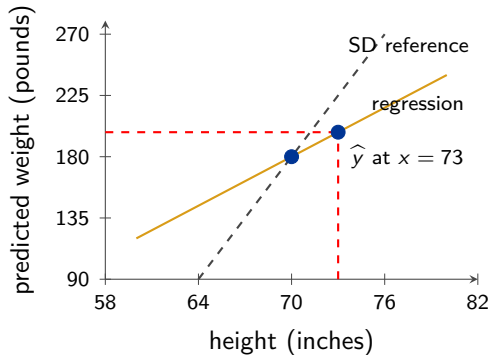
This movement can occur without an intervention.

Regression Line and Standard-Units Reference

Simple linear regression uses

$$\hat{y} = b_0 + b_1x.$$

- The **regression line** estimates average y at each x .
- The dashed **SD reference line** is a comparison line: one SD above average in x lines up with one SD above average in y .
- Unless $|r| = 1$, the regression line is pulled toward the overall average.



Standard Units

Fitted line

$$\hat{y} = \bar{y} + r \frac{SD_y}{SD_x} (x - \bar{x})$$

The prediction is an **average** y for cases with that x , not a guarantee for an individual.

Height–weight example

Average height	70 in
SD of height	3 in
Average weight	180 lb
SD of weight	45 lb
Correlation	$r = 0.40$

For $x = 73$ in, $z_x = (73 - 70)/3 = 1$.
Therefore,

$$\hat{y} = 180 + 0.40(45) = 198 \text{ lb.}$$

Optional Check: Tutoring After Low Scores

Students with unusually low midterm scores receive tutoring. On a comparable final exam, their scores improve relative to the class.

Does this prove that tutoring caused the improvement?

What else could explain part of the change? What comparison would be more informative?